Evaluating the multiple sequence alignment module of machine-learning-based protein-ligand binding prediction models

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Contents

Abstract 2
Declaration 3
Intellectual property statement 3
Acknowledgements 4
Introduction 4
Methods 9
Results 13
Discussion and Conclusion 29
References 36

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Abstract

Accurate prediction of ligand binding has emerged as a key challenge for machine learning applications in protein science. Here, we evaluate two accessible models for protein-ligand binding prediction, Chai-1 and Boltz-2, on insect odorant-binding proteins (OBPs), a structurally conserved yet sequentially diverse family. OBPs provide stringent benchmarks while binding a variety of ligands, allowing for the assessment of multiple sequence alignments (MSAs) and templates in low-homology contexts. Binding predictions were compared against crystallised structures and assessed using native and non-native ligands, including chemically and sterically implausible binders. Predicted protein backbone folds incorporating MSAs and templates closely align with experimental structures for both models, but ligand poses and interacting residues diverge significantly. Boltz-2 uniquely generates affinity estimates but struggles to accurately model ligand binding, whilst Chai-1's embedded protein language model relies on structural priors for ligand-binding predictions. Neither model's binding predictions successfully differentiated between native and non-native ligands, suggesting limited architectural integration between structural prediction and ligand binding modules. This study highlights the uses and limitations of machine-learning-based approaches for ligand binding prediction, stressing the need for improved contextual modelling of ligand affinity and specificity.

Declaration

I hereby declare that except where specific reference is made to the work of others, the contents of this dissertation are original and have not been submitted in whole or in part for consideration for any other degree or qualification in this, or any other University. This dissertation is the result of my own work and includes nothing which is the outcome of work done in collaboration, except where specifically indicated in the text. This dissertation contains less than 9000 words and has 8 or fewer figures.

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Introduction

Proteins frequently interact with a diverse series of ligands, including peptides, nucleic acids, and pheromones, to perform their functions across a wide range of cellular processes. Accurate prediction and characterisation of protein-ligand binding, rather than protein structure alone, is an essential area of study for fields including drug discovery, functional annotation, and ligand specificity [1-3]. Accurate protein-ligand binding prediction requires reliable structural characterisation of the protein backbone from primary sequence through to quaternary structure [4]. Understanding interactions between the protein backbone and ligand offers profound insights into various aspects across biology, including drug design, evolutionary insights, and biotechnological applications [5-10].

X-ray crystallography and nuclear magnetic resonance (NMR) Spectroscopy are traditionally used for structural determination of protein-ligand binding at near-atomic resolutions, with uses for high-resolution static structures and insights into smaller soluble protein structure and dynamics, respectively [11-14]. Despite refinements to crystallisation protocols and spectral determination tool development, both methods are time-consuming, costly, and highly dependent on individual protein characteristics [15, 16]. Computational approaches, like homology modelling, aim to bridge the gap between accurate structural characterisation and predictive speed, providing models which can be further applied to binding studies. However, their performance is limited by the availability of homologous templates and sequence alignments to experimentally derived structures [17]. Recent advances in machine learning have marked a transformative shift in improving the accuracy and accessibility of protein structure prediction, with the release of AlphaFold2 marking a distinct pivot in structural prediction, achieving near-experimental resolutions for community-defined protein test sets [18-21]. More recently, machine-learning models have expanded their focus towards accurately modelling protein-ligand binding [22].

Homology modelling

X-ray crystallography and NMR provide high-resolution detail for protein-ligand binding, but the number of experimentally determined structures remains far smaller than the number of known protein sequences and ligands [23]. Homology modelling addresses

this gap by assuming protein structure is more conserved than sequence; therefore, structural changes occur more slowly than sequence divergence, and similar sequences adopt comparable folds [24-26]. Traditional tools, like SWISS-MODEL and I-TASSER, rely on structural templates from the Protein Data Bank (PDB) and MSAs to guide conserved residue identification and model generation [27-31]. Homology modelling is widely applied to bridge the gap between sequence and structure, like exploring structure-function relationships of GPCR receptors [32, 33]. While effective when high-quality templates and deep MSAs are available, performance declines for poorly characterised protein families like OBPs, where evolutionary context is sparse [34-40].

Recent machine-learning approaches extend beyond homology modelling by learning general sequence-structure patterns from large datasets, enabling prediction beyond close homologues. This represents a shift from template dependency to structural inference, though challenges remain in accurately modelling ligand binding and specificity [18-21, 41, 42].

Machine learning methods for protein structure prediction

Accurate backbone prediction is essential for reliable protein-ligand modelling, as the location and associated residues of the binding pocket depend directly on the underlying fold. Protein folding involves rearranging a linear amino acid sequence into a stable, low-energy, three-dimensional conformation. Traditional predictive tools use energy functions, co-evolution, and statistical reasoning algorithms to predict conformations; however, the number of possible peptide bond conformations makes prediction from sequence alone challenging [43].

Machine learning approaches such as AlphaFold and RoseTTAFold address this by combining neural networks with transformer architecture to capture long-range sequence dependencies. They incorporate MSAs and homologous structural templates to improve accuracy and provide evolutionary context, allowing the model to infer residue-residue contacts and folding patterns [18, 22, 41-45]. Whilst this has improved backbone accuracy, it creates issues for protein-ligand binding prediction. Over-reliance on MSA transformers may bias models towards conserved features, leading to hallucinated binding sites, implausible binding poses, or a reversion to known structural priors under stress conditions [46, 47]. This is a particularly acute problem for diverse binders, orphan proteins and poorly characterised protein families such as OBPs, where evolutionary context is sparse.

Recent models attempt to reduce dependence on contextual information and de-emphasise the MSA module, training on large sequence-structure datasets, and using attention mechanisms to infer structural constraints from sequence alone. This shift

improves applicability in low-homology or MSA-free settings, though reliable prediction of ligand binding and specificity remains a major limitation without contextual information [48, 49].

Model architecture

Transformers are a neural network architecture that uses “attention” mechanisms to learn which parts of an input are most relevant to others [52]. Here, inputs typically include the target sequence, an MSA, and a structural template. AlphaFold2 popularised this approach, with subsequent models adopting a similar contextual framework [18]. These inputs are distilled into two key representations: the MSA representation and the pair representation [53].

The pair representation encodes residue-residue relationships as a matrix, with each element describing the interaction potential between residues “*i*” and “*j*” in the input sequence. The MSA module refines model predictions through two complementary attention operations: row-wise gated self-attention (modelling correlation across residues *within* a sequence) and column-wise gated self-attention (modelling conservation *across* aligned positions in homologous sequences). Combined, updates flow between the MSA and pair representation in iterative cycles, allowing long-range dependencies and co-evolutionary constraints to inform structural predictions and progressively refine structural predictions.

MSA incorporation allows models to capture co-evolutionary similarity between residues across homologs to determine the optimal folded structure of a protein. However, predictive accuracy is heavily dependent on MSA depth and quality, like homology modelling. Shallow MSA depth may reduce accuracy, with models defaulting towards known priors learnt from training data and biasing predictions towards common folds [29, 54, 55]. For poorly characterised proteins, flexible or disordered regions may be misplaced or modelled incorrectly, indirectly creating or obscuring binding pockets [56].

These limitations are exacerbated for protein-ligand prediction. MSAs shaping the backbone fold indirectly influence binding pocket formation through coevolutionary signals, with models defaulting to canonical binding pockets and pose, even when sterically or chemically implausible [57]. Shallow or absent MSAs amplify this effect; loops may collapse or be misplaced, pockets missed, and ligand placement may default to prior training-set data. For diverse protein families, like OBPs, this can lead to hallucinated binding pockets or implausible ligand binding.

Insect odorant binding proteins

Insect Odorant Binding Proteins (OBPs) are small, soluble proteins (10-30 kDa), which transport hydrophobic odorants such as pheromones through the aqueous sensillar lymph to olfactory receptor neurones, mediating mating and attraction behaviour [58–65]. Despite low sequence homology (typically ≤40%) between insect orders and species, they are structurally conserved: 6 to 8 α-helices are connected by extended loops and stabilised by six conserved cysteines forming three disulphide bonds [66-69]. This fold creates a central, hydrophobic binding pocket for the odorant ligand [70, 71]. This diversity forms a structural family with a highly conserved helical core, with variation accompanying functional diversity and evolutionary distance.

OBPs bind a wide range of ligands, with affinities spanning low millimolar to nanomolar, with exceptions extending to femtomolar [72]. Structural flexibility and the tolerance of the hydrophobic cavity enable promiscuous binding, allowing single OBPs to bind diverse and varied sets of ligands ranging from small alcohol molecules to long-chain pheromones and aromatic compounds [73-79]. Subclassifications of OBPs include PBPs (pheromone binding proteins), GOBPs (general odorant binding proteins), and LUSH (abnormal attraction to high concentrations of alcohols) [80, 81].

Here, two accessible machine-learning-based methods were selected to model known OBPs: Chai-1 and Boltz-2. Both are accessible via web servers or lightweight inference modes, without ligand input restrictions present in other models like AlphaFold3 [22].

Chai-1

Chai-1 is a machine-learning model for protein-ligand prediction that uses a multimodal architecture to integrate sequence embeddings with structural inputs [82]. Released in 2024 by Chai Discovery, it can incorporate experimental restraints, MSAs, and templates, but can also be run in single-sequence mode without external information. With model architecture developed from AlphaFold3, Chai-1 introduces residue-embedding from a large protein language model to enable strong single-sequence capabilities [22]. These embeddings capture contextual information from billions of amino acid combinations and millions of sequences, encoding biochemical properties, structural tendencies, and remote homology to construct protein backbones [83-85]. Additional features address pocket, contact, and docking constraints to capture protein-ligand interactions.

Chai-1 is distributed under open-source licensing as a Python package with high computational requirements (~2 TB, data centre-grade GPUs) or via a standalone web server.

Boltz-2

Boltz-2 is a machine-learning model for protein-ligand structure and affinity prediction, released in 2024 as a collaboration between Recursion and the MIT Jameel Clinic [86]. Building upon Boltz-1 architecture derived from AlphaFold2, it integrates protein-ligand distance constraints, method conditioning (e.g., X-ray crystallography, NMR), and multi-chain template alignment. Uniquely, it also features an affinity module for free-energy perturbation (FEP) to estimate free-energy differences between thermodynamic states, enabling binding affinity and ligand binding probability calculation as well as decoy binder discrimination [87, 88].

Boltz-2 introduces Boltz-steering, an inference-time method that applies physics-based potentials to improve the steric plausibility of predicted complexes [22, 86, 89]. For affinity prediction, a dual-head affinity module is applied downstream of the template and MSA modules, one head predicts binding likelihood while the other regresses continuous affinity values. This integrates protein-ligand structure and affinity.

Boltz-2 is distributed as a Python package with publicly accessible model weights and inference code, licensed for academic and commercial use. It is less computationally intense than Chai-1, capable of running on consumer-grade GPUs, with templates and MSAs supplied directly or from external servers.

Both models share similar architecture, with the key differences being that Boltz-2 incorporates Boltz-steering and an affinity module, while Chai-1 integrates residue-level embedding from large protein language models to improve single-sequence capabilities and de-emphasise MSA module reliance [18-20, 53-55].

Research Gap

Few studies have evaluated machine-learning methods for protein-ligand predictions on poorly characterised protein families and non-canonical ligands. Boltz-2 and Chai-1 are open-source models providing code and model weights for academic and commercial use, but their robustness in low-data environments, with poorly conserved protein families, or against chemically and sterically implausible ligands is unexplored. Here, we use the sequentially diverse and structurally conserved OBP family as a test set.

We compare protein-ligand structural predictions generated with and without MSA and template input to assess their impact upon model accuracy. We further challenge the models with chemically and sterically implausible non-native ligands to determine how they determine binding plausibility with and without evolutionary context. We hypothesise that both models will accurately predict backbone structure and ligand binding when MSAs and templates are provided, but only Chai-1 will maintain backbone

accuracy without this input due to its incorporation of residue-level embeddings from a large protein language model. We expect both models will hallucinate binding sites for implausible ligands to accommodate binding on top of the existing protein fold for both conditions, highlighting weaknesses in model architecture and layout.

This study provides a direct comparison between Boltz-2 and Chai-1 using sequentially diverse proteins with low homology, poor conservation, and shallow MSAs, whilst comparing their treatment of sterically and biologically implausible ligands.

Methods

Odorant Binding Protein Selection

X-ray crystallographic data from 24 OBP complexes were retrieved from the RCSB Protein Data Bank. These included 5 unbound (apo) proteins and 19 corresponding ligand-bound complexes [90]. The selection criteria required high-resolution unbound structures and at least two co-crystallised, ligand-bound states per protein. Native structure files (.pdb/.cif) and corresponding sequences (.fasta) were downloaded. Unbound (apo) PDB structures are indicated in italics:

- General Odorant Binding Protein 2 (GOBP2) from *Bombyx mori* (PDB: *2WCK*, 2WCL, 2WCJ, 2WCM, 2WCH) [76]
- LUSH from *Drosophila melanogaster* (PDB: *1T14*, 1OOF, 1OOG, 1OOH, 2GTE) [73]
- Odorant Binding Protein 22 (OBP22) from *Aedes aegypti* (*6OG0*, 6OPB, 6OII, 6OTL, 6OMW, 6OGH) [74]
- Pheromone Binding Protein ASP1 from *Apis mellifera* (PDB: *2H8V*, 3BFH, 3BFB, 3BFA, 3FE9) [75]
- Odorant Binding Protein 22a (OBP22a) from *Anopheles gambiae* (PDB: *3L47*, 3L4L, 3QME) [77-79]

Ligand Tables

Ligand tables for OBPs were compiled from literature sources and the Protein Data Bank (PDB) [73-79]. SMILES representations for each ligand were obtained from PubChem, and 2D images were generated using the Novoprolabs Smi2SVG tool [91]. Tables provide contextual information for each OBP, including species, ligand identity, and chemical structure.

Multiple Sequence Alignment

The five OBP sequences (.fasta) were aligned using CLUSTALW in MEGA12 [92, 93]. Pairwise and multiple alignments were generated with a gap opening penalty of 10 and a gap extension penalty of 2, and the BLOSUM protein weight matrix was applied [94]. Additional parameters, including residue-specific penalties, hydrophilic penalties, end gap separation, and a negative matrix option, were enabled. A gap separation distance of 4 and a 30% delay divergent cutoff was applied.

Evolutionary Analysis

Phylogenetic analysis of the 5 OBPs was performed in MEGA12 using the Maximum Likelihood (ML) method with the Jones-Taylor-Thornton (JTT) amino acid substitution model [93, 95, 96]. 1000 bootstrap replicates assessed node support [97]. The initial tree was chosen based on the highest log-likelihood ratio (-1,748.57) between a Neighbour-Joining (NJ) tree and a Maximum Parsimony (MP) tree [98, 99]. Among ten replicate MP searches from random starting trees, the selected tree had the shortest length.

Topology Map

PDBsum generated a topological map of the example OBP (*Bombyx mori* GOBP2 (apo form)) [100]. N- and C-termini were labelled, α-helices and corresponding residue ranges are shown.

Boltz-2

Boltz-2 was run locally on a Windows workstation with an NVIDIA GTX 4060 [86]. A BASH script and YAML configuration files were used to generate two predictions per protein-ligand complex: one with MSA and structural templates, and one without. MSAs were generated with MMseqs2 database without subsampling. MMseqs2 generates alignments using the Uniref Colabfold database, which integrates alignments from BFD, MGnify, MetaEuk, and SMAG [101]. Structural templates were ligand-free (apo) protein structures in .cif format from the PDB. Key interacting residues from crystallographic data and literature were used to refine template selection. Ligand information was provided as SMILES strings.

Predictions used 10 recycling steps and 25 diffusion sampling steps, mimicking AlphaFold3 defaults. The MSA-enabled pipeline generated alignments using MMseqs2, while in the non-MSA pipeline, the MSA field was omitted [101]. Full predicted aligned error (PAE) and predicted distance error (PDE) matrices are enabled for both, and the

output is set to the .pdb file format. Predicted TM-scores (pTM) to intrachain accuracy and inter-protein TM-score (ipTM) for interchain accuracy were reported as default [102].

Boltz-2 produced affinity estimates using its free-energy perturbation (FEP) module [103, 104]. Binding affinities were reported as log(IC_{50}) values (μM), converted to IC_{50} (10^log(IC_{50})) for interpretation. A binary binder probability score (0-1) was generated. Both metrics were compared across MSA and non-MSA predictions.

Chai-1

Chai-1 was run via its web interface, as local installation required high storage requirements (>2 TB) and data-centre grade GPUs (NVIDIA A100/H100) [82]. Two predictions were generated for each protein-ligand complex: one with MSA and structural templates, and one without.

MSAs were generated using MMSeqs2 [101]. Wherever possible, Chai-1 retrieved precomputed MSAs from the OpenProteinSet database [105]. Structural templates were obtained from the PDB, and ligands were provided as SMILES strings.

Predicted structures were returned in .cif format and converted to .pdb files using OpenBabel [106]. Chai-1 outputs included predicted TM-scores (pTM) for intrachain accuracy and inter-protein TM-scores (ipTM) for interchain accuracy [86]. Chai-1 does not output affinity scores or binder probability scores.

PyMOL

Experimentally determined (PDB) and predicted structures were analysed using PyMOL [90, 107]. For each complex, residues within 4Å of the ligand were identified, a standard cut-off for non-covalent interactions, such as hydrogen bonds, van der Waals contacts, and π-π stacking. Structural images were generated using the capture function with ray tracing enabled and exported for figure preparation.

FATCAT Alignment

Protein similarity was quantified using FATCAT, calculating RMSD (Å) of Cα atomic coordinates [108]. FATCAT optimises alignments and minimises rigid body movements around pivot points. Canonical PDB structures were aligned with both MSA and non-MSA Chai-1 and Boltz-2 predictions to assess structural similarity.

Template Modelling Scores

TM-scores were used to quantify fold similarity between MSA and non-MSA Chai-1 and Boltz-2 predictions, with 1 indicating a perfect match. TM-scores were calculated using the Zhang Lab TM-score program [109, 110].

Kinemage Analysis

To test whether models could distinguish biologically plausible and implausible ligands, non-native ligands, chemically or structurally implausible, were included as negative controls. A protein BLAST against *Bombyx mori* GOBP2 identified sequences with high query coverage but low sequence identity. Discontiguous megablast optimised search sensitivity for distantly related sequences [111, 112].

Bombyx mori GOBP2 was chosen as a reference due to its well-characterised hydrophobic binding pocket, where residues 33–35 restrict back-pocket access and limit Phe41 displacement [76]. Boltz-2 and Chai-1 generate binding predictions for all ligands (Table 2), testing structurally and chemically implausible protein-ligand binding.

Predicted complexes were visualised in PyMOL to assess binding poses. Van der Waals contacts were analysed using the MolProbity online server, with hydrogen atoms added via reduce, and sidechain flips for Asn, Gln, and His residues applied where appropriate [113, 114]. All-atom contacts were analysed using default settings to generate 3D kinemages, visualised in KiNG [115]. Cα atoms are represented in white, side chains in blue. Van der Waals contacts between side chains and ligands are represented as blue/green dots, and unfavourable overlaps are highlighted in red/yellow. Kinemage outputs were exported as .jpg files for annotation and comparison.

In Silico Mutated Odorant Binding Protein

Bombyx mori GOBP2 (PDB: 2WCK) was mutated in silico, replacing residues within 4Å of native ligands with non-polar glycine to remove side-chain mediated interactions while preserving the canonical OBP fold. Conserved cysteine residues forming disulphide bonds were retained to preserve the original fold. Hexadecane, a non-polar ligand biologically implausible for canonical contacts, was used to assess the model's predictive binding with limited evolutionary context. Predictions were generated with the Boltz-2 and Chai-1 pipelines described previously.

Results

Structural, Evolutionary, and Phylogenetic Analysis of OBPs

Five OBPs were selected from the PDB based on the availability of apo and ligand-bound crystal structures. CLUSTALW alignment revealed limited sequence conservation, with protein lengths ranging from ~120 to ~170 amino acids (Fig. 1a) [92, 93]. Extended regions of conservation (>5 residues) were rare; an example being 65-70 in ASP1 and GOBP2. Negatively charged residues such as Glu110 and Glu/Asp36 showed partial conservation, but only Glu110 and six cysteine residues (Cys24, Cys56, Cys60, Cys119, Cys132, and Cys141) were fully conserved across OBPs. This is characteristic of OBPs, where disulphide bond-forming cysteines are essential for maintaining the canonical fold and binding pocket.

Phylogenetic analysis, performed using the Maximum Likelihood method and the Jones-Taylor-Thornton model, produced branch lengths representing genetic distance (Fig. 1b) [93, 95, 96]. Mosquito OBPs (*Aedes aegypti* OBP22 and *Anopheles gambiae* OBP22a) formed a distinct clade, *Apis mellifera* APS1 and *Bombyx mori* GOBP2 clustered together, while *Drosophila melanogaster* LUSH did not cluster. Bootstrap support was high for the ASP1-GOBP2 cluster (88) but was lower for OBP22-OBP22a (52), reflecting uncertainty in the evolutionary placement. This may reflect differences in ligand specificity, with mosquito OBPs showing weaker conservation and lower phylogenetic confidence compared to ASP1 and GOBP2.

The topology and structure of *Bombyx mori* GOBP2 illustrate the conserved OBP fold, with eight α-helices (Fig. 1c, d). In GOBP2, the conserved disulphide bonds between cysteine residues link Cys19-Cys54 (α2 and α4), Cys50-Cys108 (α4 and α7), and Cys97-Cys117 (α6 and α7). These reflect the canonical OBP structure of six to eight α-helices. These disulphide bridges form the central binding pocket, reflecting how OBPs form a conserved, stable core framework despite limited sequence conservation. This stability likely underpins the broad ligand-binding capability of OBPs.

Native Ligands for Selected Odorant Binding Proteins

The native ligands of the selected OBPs include pheromones with long-chain alkyl groups and functional groups such as carboxyl, aldehyde, ketone, alcohol or sulphonamide groups (Table 1). Some ligands contain benzene rings or cyclic alkanes, while smaller alcohols (ethanol, propanol, butanol) are represented. The chemical diversity of ligands highlights the binding diversity of OBPs, while the presence of long alkyl chains suggests a conserved structural feature for ligand recognition.

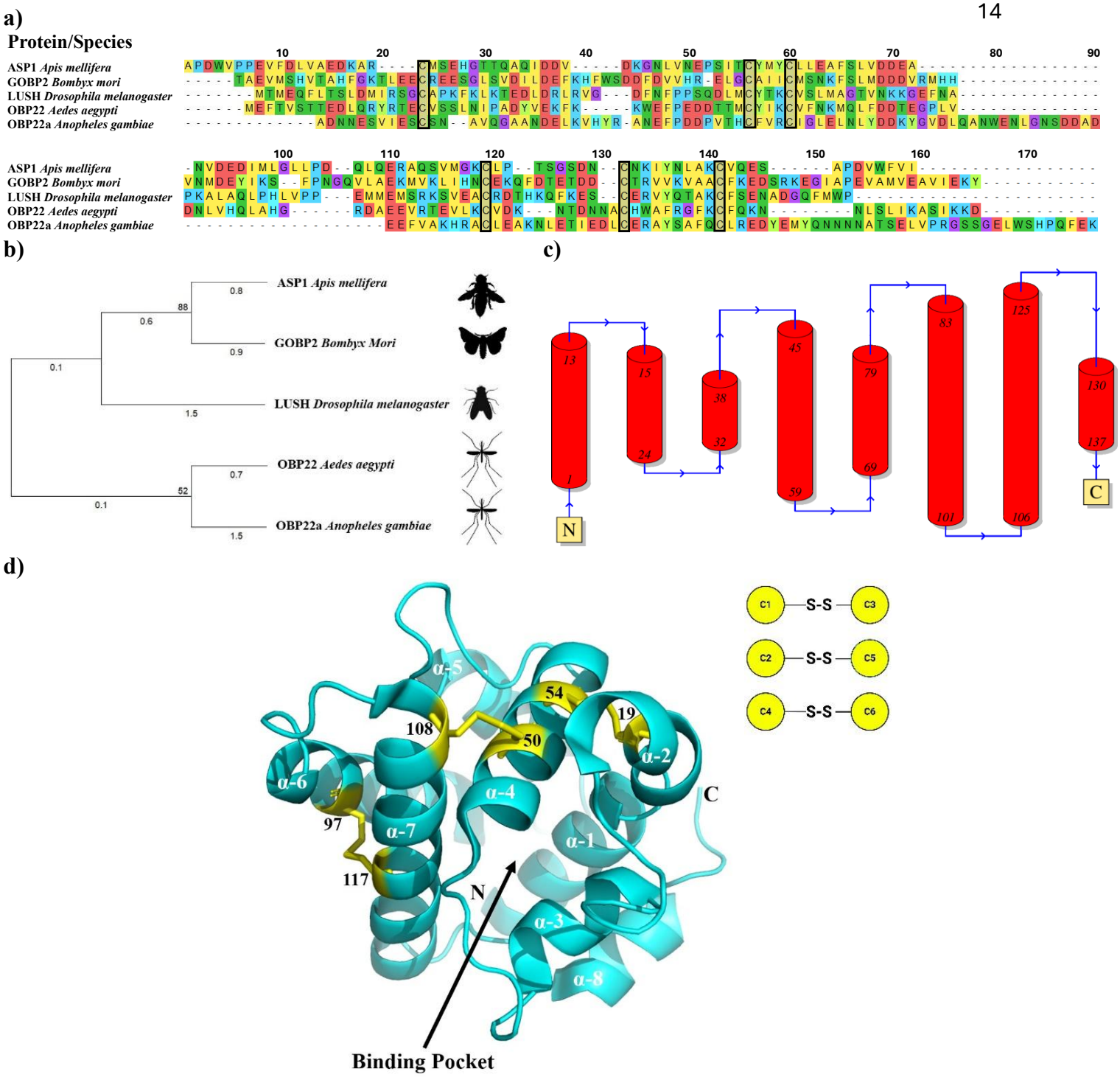
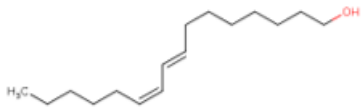
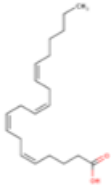
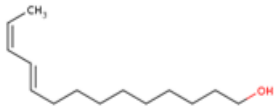
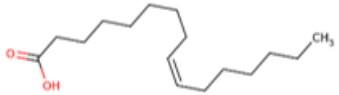
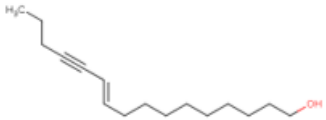
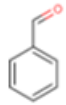
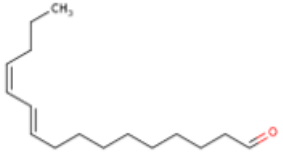
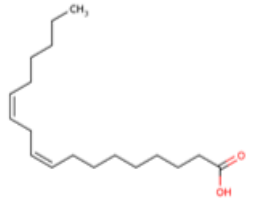
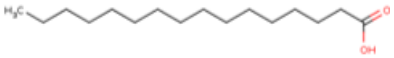
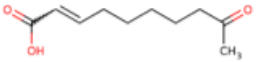
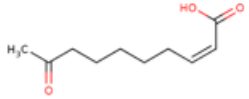
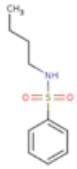
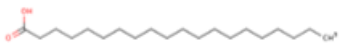


Figure 1. Structural, evolutionary, and phylogenetic analysis of Odorant Binding Proteins (OBPs). Insect OBPs were selected based on their structural homology, low sequence conservation, and diverse ligand binding properties. Five OBPs were selected from the Protein Data Bank based on the availability of apo and ligand-bound crystal structures. **(a)** CLUSTALW Multiple Sequence Alignment of five selected OBPs; conserved cysteine residues are indicated. **(b)** Rooted Phylogenetic Tree inferred using Maximum Likelihood. **(c)** Topology map of *Bombyx mori* GOBP2 (apo form). N- and C- termini are labelled; α-helices and corresponding residue ranges are shown. **(d)** 3D structure of *Bombyx mori* GOBP2. N- and C-termini are labelled; α-helices (α-1–α-8) are indicated. Disulphide bridges between cysteine residues are highlighted in yellow, with positions labelled. The canonical, experimentally determined binding pocket is shown.

Table 1. Native Ligands for Selected Odorant Binding Proteins

	Organism	OBP	PDB	Ligand	Structural Formula		Organism	OBP	PDB	Ligand	Structural Formula
1	*Bombyx mori*	GOBP2	2WCL	(8E, 10Z)-hexadecadien-1-ol		10	*Aedes aegypti*	OBP22	6OII	Arachidonic acid	
2	*Bombyx mori*	GOBP2	2WCJ	(10E,12Z)-tetradecadien-1-ol		11	*Aedes aegypti*	OBP22	6OMW	Palmitoleic acid	
3	*Bombyx mori*	GOBP2	2WCM	(10E)-hexadecen-12-yn-1-ol		12	*Aedes aegypti, Anopheles gambiae*	OBP22, OBP22a	6OTL, 3L4L	Benzaldehyde	
4	*Bombyx mori*	GOBP2	2WCH	Bombykal		13	*Aedes aegypti*	OBP22	6OGH	Linoleic acid	
5	*Drosophila melanogaster*	LUSH	1OOF	Ethanol		14	*Apis mellifera*	ASP1	3BFH	Hexadecanoic acid	
6	*Drosophila melanogaster*	LUSH	1OOG	Propanol		15	*Apis mellifera*	ASP1	3BFB	9-keto-2(E)-decenoic acid	
7	*Drosophila melanogaster*	LUSH	1OOH	Butanol		16	*Apis mellifera*	ASP1	3BFA	(2Z)-9-oxo-dec-2-enoic-acid	
8	*Drosophila melanogaster*	LUSH	2GTE	11-cis-vaccenyl acetate		17	*Apis mellifera*	ASP1	3FE9	*n*-Butyl benzene-sulphonamide	
9	*Aedes aegypti*	OBP22	6OPB	Arachidic acid		18	*Anopheles gambiae*	OBP22a	3QME	Cyclohexanone	

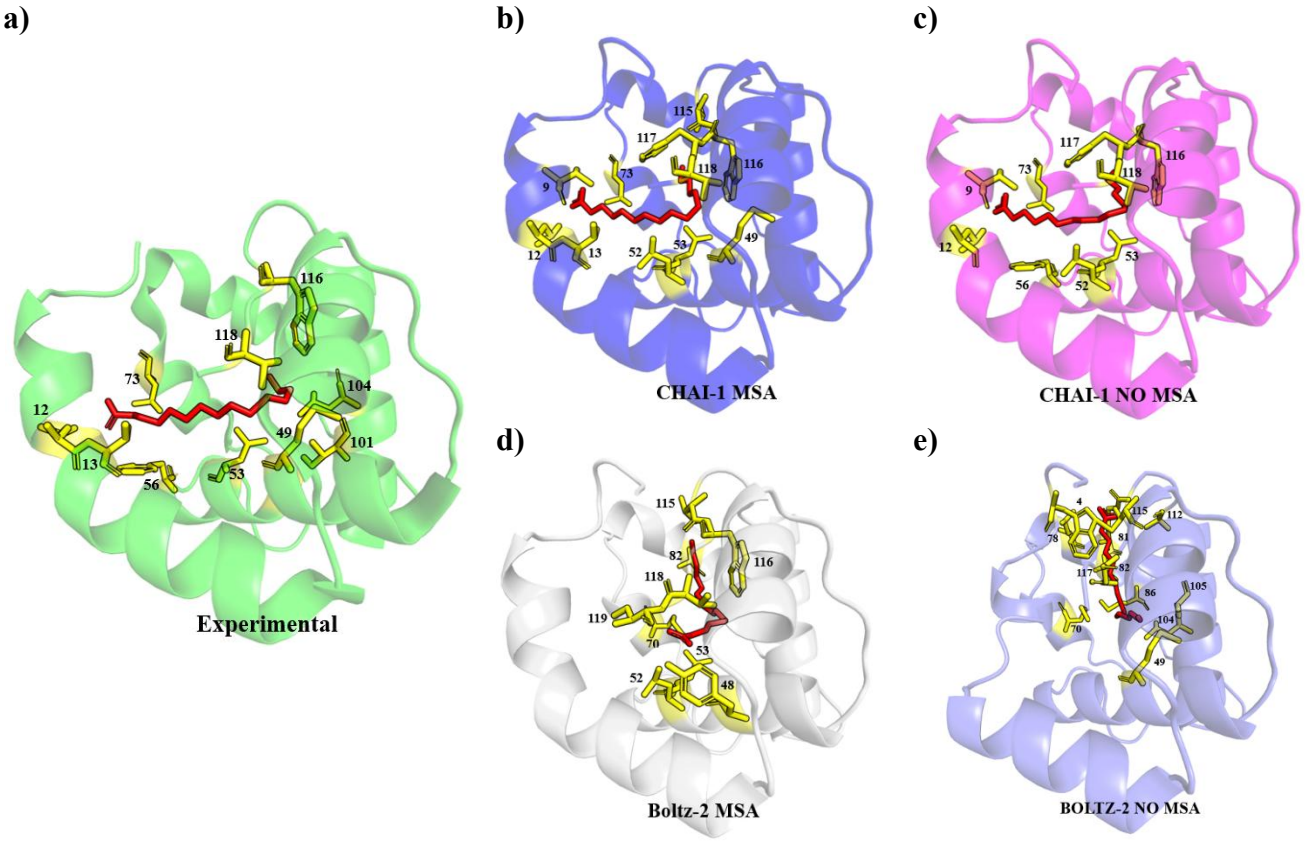
Structural predictions of *Apis mellifera* Pheromone-Binding Protein (ASP1) bound to hexadecanoic acid

The canonical PDB structure of ASP1 with its native ligand hexadecanoic acid shows the ligand occupying the central hydrophobic pocket (Fig. 2a). Key interacting residues within 4Å include non-polar residues with hydrophobic side chains, with potential hydrogen bonds forming between the ligand's carboxyl group and the backbone amides of Leu12, Val13, and Leu73. Aromatic and aliphatic residues, like Phe56, likely contribute to van der Waals contacts along the ligand's alkyl chain. These canonical interactions provide a reference for model comparison.

With or without MSA, Chai-1 closely replicates the canonical ligand pose and backbone fold (Fig. 2a, b). The MSA model introduces an additional Val9 contact within 4Å of the carboxyl group, while subtle shifts in the ligand tail alter which residues are classified as interacting (Fig. 2a, b, f). The non-MSA model retains canonical contacts like Phe56, but loses others, slightly reducing the binding interface, potentially due to ligand tail repositioning (Fig. 2c). These modest deviations likely reflect the challenges in modelling flexible ligand regions or alternative low-energy conformations of the flexible alkyl tail. These subtle differences suggest Chai-1 captures ligand pose and interactions, with carboxyl group orientation and backbone fold retained.

In contrast, Boltz-2 differs substantially from Chai-1 and canonical predictions (Fig. 2d, e). The MSA model repositions the carboxyl towards the pocket edge, introducing unique contacts (e.g., Tyr48), with few overlapping residues (Leu53, Trp116, Val118) to the canonical structure (Fig. 2d, f). The non-MSA prediction deviates further, pushing the ligand to the pocket periphery, forming contacts with multiple alanine residues (Fig. 2e, f). This retains the canonical α-helices but differs significantly in the backbone fold.

All predictions restrict interacting residues to α-helices, consistent with the conserved fold of OBPs (Fig. 2a-e). Combined, these results suggest Chai-1 captures ligand pose and contacts with reasonable accuracy, whilst Boltz-2 appears less effective at modelling native ligand poses, favouring residue–ligand chemical compatibility over preserving the canonical binding arrangement. The inclusion of MSA and template information plays a key role in retaining the backbone fold.



f)

	0 10 20 30 40 50 60
ASP1 (3BFH – PDB)	APDWVPPEVFDLVAEDKARCMSEHGTTQAQIDDVDKGNLVNEPSITCYMYCLLEAFSLVD
Boltz-2 MSA/TEMPLATE	APDWVPPEVFDLVAEDKARCMSEHGTTQAQIDDVDKGNLVNEPSITCYMYCLLEAFSLVD
Boltz-2 NO MSA/TEMPLATE	APDWVPPEVFDLVAEDKARCMSEHGTTQAQIDDVDKGNLVNEPSITCYMYCLLEAFSLVD
Chai-1 NO MSA/TEMPLATE	APDWVPPEVFDLVAEDKARCMSEHGTTQAQIDDVDKGNLVNEPSITCYMYCLLEAFSLVD
Chai-1 MSA/TEMPLATE	APDWVPPEVFDLVAEDKARCMSEHGTTQAQIDDVDKGNLVNEPSITCYMYCLLEAFSLVD

	70 80 90 100 110
3BFH – PDB	DEANVDEDIMLGLLPDQLQERAQSVMGKCLPTSGSDNCNKIYNLAKCVQESAPDVWFVI
Boltz-2 MSA/TEMPLATE	DEANVDEDIMLGLLPDQLQERAQSVMGKCLPTSGSDNCNKIYNLAKCVQESAPDVWFVI
Boltz-2 NO MSA/TEMPLATE	DEANVDEDIMLGLLPDQLQERAQSVMGKCLPTSGSDNCNKIYNLAKCVQESAPDVWFVI
Chai-1 NO MSA/TEMPLATE	DEANVDEDIMLGLLPDQLQERAQSVMGKCLPTSGSDNCNKIYNLAKCVQESAPDVWFVI
Chai-1 MSA/TEMPLATE	DEANVDEDIMLGLLPDQLQERAQSVMGKCLPTSGSDNCNKIYNLAKCVQESAPDVWFVI

Figure 2. Structural predictions of *Apis mellifera* Pheromone-Binding Protein (ASP1) bound to the native ligand hexadecanoic acid (PDB ID: 3BFH). Hexadecanoic acid is highlighted in red, and its predicted binding pose within the predicted binding pocket is shown. Interacting residues within 4Å of the ligand are highlighted in yellow and labelled. **(a)** Canonical PDB structure of ASP1 with Hexadecanoic acid. **(b)** Boltz-2 predicted with MSA and template input. **(c)** Boltz-2 prediction without MSA or template. **(d)** Chai-1 predicted with MSA and template input. **(e)** Chai-1 predicted without MSA or template. **(f)** ASP1 sequence alignment with interacting residues underlined and highlighted in yellow.

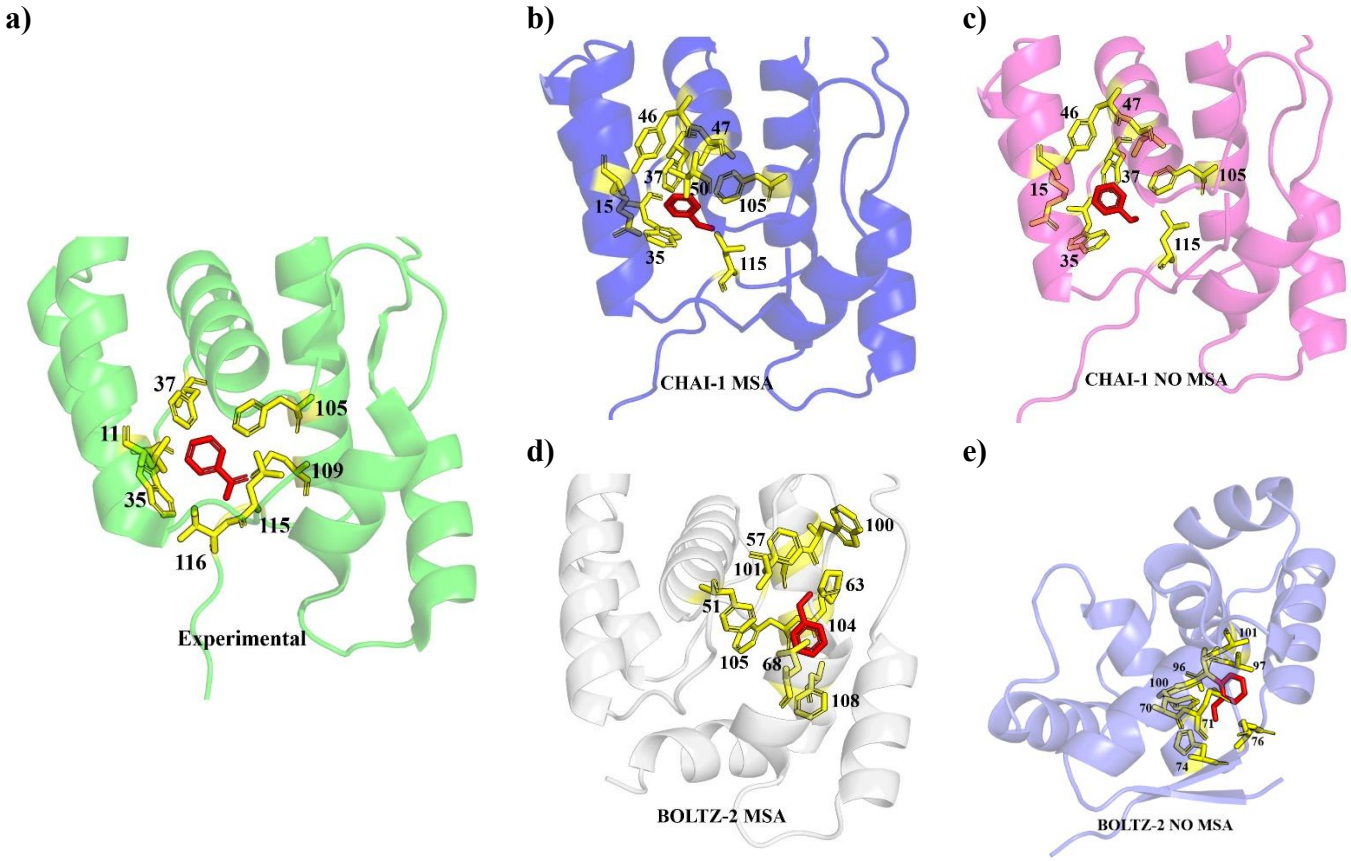
Structural predictions of *Aedes aegypti* Odorant Binding Protein (OBP22) bound to benzaldehyde

The canonical PDB structure of OBP22 with its native ligand benzaldehyde shows the ligand binding deep within the central hydrophobic pocket (Fig. 3a). Key interacting residues within 4Å include amino acids with hydrophobic side chains and aromatic contacts, while the polar residue Gln109 lies within hydrogen-bonding distance of the aldehyde, suggesting a mixed hydrophobic-polar binding mode.

With or without MSA, Chai-1 closely replicates the canonical backbone with subtle variations in pose (Fig. 3a, b, c). The MSA model introduces additional interactions (Tyr46, Ile47, and Val50) and loses others (Lys11, Gln109, and Ile116), potentially expanding the hydrophobic and aromatic binding interface (Fig. 3b). The non-MSA model shifts benzaldehyde slightly downward within the pocket but, like the MSA model, retains key contacts (Trp35, Phe37, Phe105, Leu115) (Fig. 3b, f). This suggests the importance of these key hydrophobic and aromatic residues, with these large residues likely stabilising benzaldehyde ligand binding due to its small size.

In contrast, Boltz-2 diverges from Chai-1 and the canonical structure (Fig. 3d, e). The MSA model maintains the α-helical fold, but relocates benzaldehyde from the canonical pocket, resulting in minimal overlap of interacting residues, with Phe105 the only conserved contact (Fig. 3f). Hydrophobic interacting residues are retained in the new site, likely to interact with the aromatic or alkyl portions of benzaldehyde (Fig. 3d). Additionally, the arrangement of α-helices is different from that of OBP22. The non-MSA model shows pronounced deviations, including formation of a β-sheet not present in OBPs and a relocated binding site dominated by polar and charged residues, contrasting with the hydrophobic environment of the canonical pocket (Fig. 3e). This indicates that template information is essential for backbone modelling for Boltz-2.

Combined, these results suggest Chai-1 reliably captures the backbone fold and core interacting residues with slight differences in ligand pose, while Boltz-2 appears less effective at modelling the binding pose and global fold, suggesting reduced reliability for modelling ligand placement of small, flexible ligands that can occupy multiple binding sites.



f)

	0 10 20 30 40 50 60
OBP22 (6OTL – PDB)	MEFTVSTTEDLQRYRTECVSSLNIPADYVEKFKKWEFPEDDTTMCYIKCVFNKMQLFDDT
Boltz-2 MSA/TEMPLATE	MEFTVSTTEDLQRYRTECVSSLNIPADYVEKFKKWEFPEDDTTMCYIKCVFNKMQLFDDT
Boltz-2 NO MSA/TEMPLATE	MEFTVSTTEDLQRYRTECVSSLNIPADYVEKFKKWEFPEDDTTMCYIKCVFNKMQLFDDT
Chai-1 MSA/TEMPLATE	MEFTVSTTEDLQRYRTECVSSLNIPADYVEKFKKWEFPEDDTTMCYIKCVFNKMQLFDDT
Chai-1 NO MSA/TEMPLATE	MEFTVSTTEDLQRYRTECVSSLNIPADYVEKFKKWEFPEDDTTMCYIKCVFNKMQLFDDT

	70 80 90 100 110 120
6OTL – PDB	EGPLVDNLVHQLAHGRDAEEVRTEVLKCVDKNTDNNACHWAFRGFKCFQKNNLSLIKASIKKD
Boltz-2 MSA/TEMPLATE	EGPLVDNLVHQLAHGRDAEEVRTEVLKCVDKNTDNNACHWAFRGFKCFQKNNLSLIKASIKKD
Boltz-2 NO MSA/TEMPLATE	EGPLVDNLVHQLAHGRDAEEVRTEVLKCVDKNTDNNACHWAFRGFKCFQKNNLSLIKASIKKD
Chai-1 MSA/TEMPLATE	EGPLVDNLVHQLAHGRDAEEVRTEVLKCVDKNTDNNACHWAFRGFKCFQKNNLSLIKASIKKD
Chai-1 NO MSA/TEMPLATE	EGPLVDNLVHQLAHGRDAEEVRTEVLKCVDKNTDNNACHWAFRGFKCFQKNNLSLIKASIKKD

Figure 3. Structural predictions of *Aedes aegypti* Odorant Binding Protein 22 (OBP22) bound to the native ligand benzaldehyde (PDB ID: 6OTL). Benzaldehyde is highlighted in red, and its predicted binding pose is shown. Interacting residues within 4Å of the ligand are highlighted in yellow and labelled. **(a)** Canonical PDB structure of OBP22 with Benzaldehyde . **(b)** Boltz-2 predicted with MSA and template input. **(c)** Boltz-2 prediction without MSA or template. **(d)** Chai-1 predicted with MSA and template input. **(e)** Chai-1 predicted without MSA or template. **(f)** 6OTL sequence alignment with interacting residues underlined and highlighted in yellow.

Accuracy of Boltz-2 and Chai-1 Predictions

The accuracy of Boltz-2 and Chai-1 predictions was evaluated by comparing predicted OBP structures to canonical PDB structures using FATCAT for backbone alignment (Fig. 4a, b) [108]. Boltz-2 predictions show clear differences between MSA-based and non-MSA-based models. MSA predictions were generally accurate (RMSD <1 Å), while non-MSA predictions often exceeded 2 Å. Exceptions (6OTL, 6OMW) still retained lower RMSDs for MSA predictions, suggesting the reliance of Boltz-2 upon contextual information for backbone accuracy. Chai-1 predictions showed consistently low RMSD values (<1 Å) for both inputs, indicating it is less dependent upon templates and MSAs for backbone predictions (Fig. 4b). For Chai-1, two predictions had higher non-MSA RMSD values than MSA inputs: *Aedes aegypti* OBP22 bound to benzaldehyde (6OMW; MSA: 2.0 Å, No MSA: 1.0 Å) and Apis mellifera ASP1 bound to N-Butylbenzenesulfonamide (3FE9; MSA: 0.9 Å, No MSA: 0.4 Å). Notably, the repeated difficulty with benzaldehyde (6OMW) likely reflects the inherent challenge of modelling small, flexible ligands with multiple potential binding poses.

TM-score analysis further highlights these trends. Boltz-2 MSA-based predictions generally achieved TM-scores >0.9, while non-MSA predictions were much lower (<0.4), indicating the importance of MSA and template data for Boltz-2 (Fig. 4c). An exception was *Apis mellifera* ASP1 (3BFH, 3BFB, 3BFA, 3FE9), where non-MSA predictions had TM-scores ≥0.8 Å, suggesting that Boltz-2 models this protein effectively regardless of contextual information. Chai-1 TM-scores were consistently high for each method (>0.9), indicating less reliance on MSA and template data (Fig. 4d). Both models struggled with *Anopheles gambiae* OBP22a (3L4L, 3QME) with low overall fold similarity (TM-score <0.2) (Fig. 4c, d). As other OBPs measured are structurally homologous, these low scores may reflect scoring metric or OBP22a-unique structural features.

Confidence scores (pTM for global structure, ipTM for inter-chain interfaces) showed that Boltz-2 MSA predictions had pTM scores >0.7 and slightly lower ipTM scores, which follow a similar pattern (Fig. 4e). Non-MSA predictions showed more variability, with pTM and ipTM values generally lower than MSA equivalents. ipTM scores were consistently lower than corresponding pTM scores, indicating inter-chain interface prediction is more challenging than backbone prediction. Chai-1 models maintained high pTM (>0.8) for both MSA and non-MSA, but ipTM scores were generally low (<0.5), indicating Chai-1 also finds predicting inter-chain interfaces challenging (Fig. 4f). MSA inputs generally exceed their equivalent non-MSA scores. Combined, this suggests Boltz-2 relies heavily on MSA and template information for modelling global structure and inter-chain interfaces, whilst Chai-1 is confident in modelling backbones without contextual information but struggles in predicting inter-chain interfaces.

Boltz-2 predicted protein-ligand binding affinities and probabilities

Predicted binding affinities for OBPs and their native ligands were generally below 50 μM, with minimal differences between predictive methods, consistent with strong native ligand binding to OBPs (Fig. 4g). Exceptions include *Aedes aegypti* OBP22 bound to benzaldehyde (6OTL), and *Apis mellifera* ASP1 bound to 9-keto-2(E)-decenoic acid (3BFB) or (2Z)-9-xodec-2-enoic-acid (3BFA). Non-MSA predictions yielded weaker affinities (e.g. 3BFA; MSA: 157μM, non-MSA: 293μM). This suggests contextual information improves binding pocket accuracy, even for ligands predicted to bind weakly. Other *Apis mellifera* ligands (hexadecanoic acid, 3BFH; N-butylbenzenesulfonamide, 3FE9) show low predicted affinities despite their biological relevance. This highlights potential limitations of Boltz-2 to model complex, induced-fit interactions. This suggests Boltz-2 may not model nuanced hydrophobic and hydrogen bond-driven interactions typical for fatty acids and aromatic ligands.

Most predicted probability binaries were <0.5, indicating Boltz-2 considered such binding interactions unlikely (Fig. 4h). No consistent pattern emerged for MSA versus non-MSA predictions. *Drosophila melanogaster* LUSH bound to simple alcohols (ethanol, propanol, and butanol [1OOF-1OOH]) has the lowest predicted probabilities, with both MSA and non-MSA inputs <0.2, despite their known biological relevance, while their more complex, hydrophobic ligand 11-cis-vaccenyl-acetates (2GTE) had a higher probability binary (~0.4). Long-chain fatty acids bound to *Aedes aegypti* OBP22 (arachidic acid (6OPB), arachidonic acid (6OII), palmitoleic acid (6OMW), linoleic acid (6OGH)) showed high predicted probabilities (>0.8) with MSA/templates. Benzaldehyde bound to *Aedes aegypti* OBP22 (6OTL) and *Anopheles gambiae* OBP22a (3L4L) yielded lower probabilities, implying ligand chemistry influences predictive outcomes, potentially capturing broad trends in hydrophobic, long-chain ligands but struggling with high-affinity binding of small, polar odorants.

Boltz-2 reliably models the *Aedes aegypti* OBP22 family of long-chain fatty acids, showing moderate correlation between low predicted affinities and high probability binaries (Fig. 4g, h). Benzaldehyde was predicted to bind weakly in both metrics, potentially highlighting limitations of Boltz-2 ligand modelling. Affinity and probability metrics did not consistently align, with both being low for the *Drosophila melanogaster* LUSH family (1OOF – 2GTE) despite biological relevance. Consequently, this suggests Boltz-2 predictions should not replace experimental binding assays for OBPs.

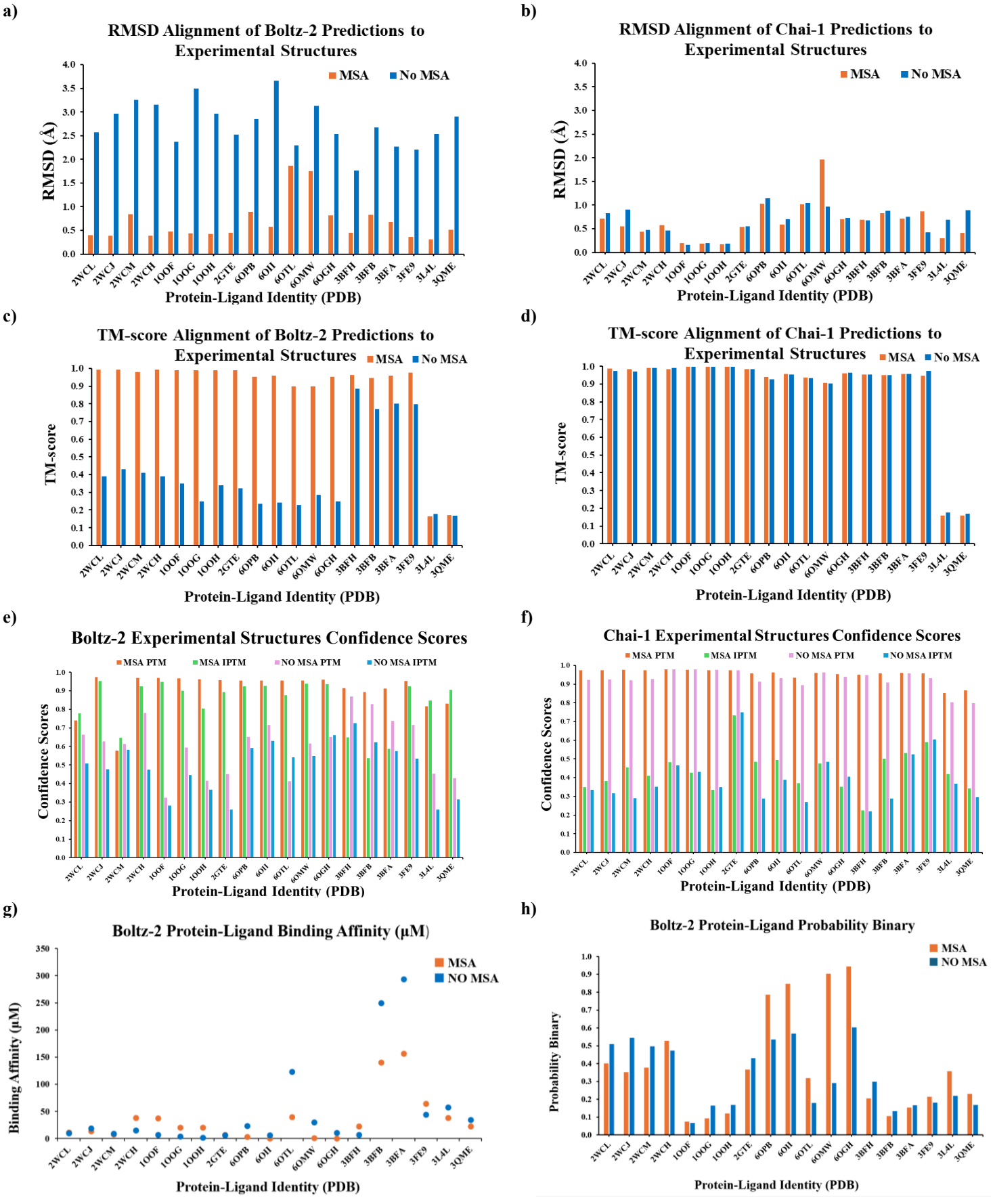


Figure 4. Comparative structural and ligand-binding performance of Boltz-2 and Chai-1 predictions. Structural similarity was evaluated relative to experimental PDB structures using RMSD (Å, Cα coordinates), TM-score (overall fold similarity), predicted TM-score (pTM, intrachain accuracy), and inter-chain TM-score (ipTM, interchain accuracy). Boltz-2 predictions also include unique ligand-binding metrics: predicted affinity (μM) and probability binary (0–1, with 1 indicating maximal confidence in binding). **(a)** RMSD of Boltz-2 predictions. **(b)** RMSD of Chai-1 predictions. **(c)** TM-score of Boltz-2 predictions. **(d)** TM-score of Chai-1 predictions. **(e)** Boltz-2 confidence scores (pTM and ipTM). **(f)** Chai-1 confidence scores (pTM and ipTM). **(g)** Boltz-2 protein-ligand predicted binding affinity (μM). **(h)** Boltz-2 protein-ligand probability binary.

Native and non-native ligands for BLAST-aligned sequences to *Apis mellifera* Pheromone Binding Protein 1 (ASP1)

Native and non-native ligands of sequences BLAST-aligned to *Apis mellifera* Pheromone Binding Protein 1 (ASP1) were recorded (Table 2). Native ligands include pheromones featuring long-chain alkyl groups and functional groups such as carboxyl, ketone, alcohol or epoxy.

Non-native ligands were included to test chemical and steric constraints. These included the straight-chain alkane, hexadecane, and polycyclic aromatic hydrocarbons, such as pyrene (four fused benzene rings) and coronene (six peri-fused benzene rings). These compounds are included as chemically and sterically implausible and serve as negative controls for binding predictions.

Predicted structures of non-native ligands to Odorant Binding Proteins

Predicted complexes of *Bombyx mori* GOBP2 with the large, rigid polycyclic ligand coronene were generated using MSA and template settings before PyMol visualisation (Fig. 5a). Both models produce protein backbones closely matching the canonical GOBP2 structure, with minor deviations at the smallest α-helix and termini. Coronene was placed within the canonical hydrophobic pocket in similar binding poses for both models.

KiNG all-atom contact analysis revealed multiple steric clashes between coronene and pocket residues in both models, indicating predicted poses are structurally implausible without significant backbone rearrangement (Fig. 5b, c). Persistent clashes were observed with Val8, Met73, and Ala115 in both models, with Phe36 uniquely overlapping in Boltz-2. Hydrophobic residues with small side chains (Ala, Val) and flexible residues like Met are incapable of accommodating the rigid coronene without steric conflict. Overlaps involving aromatic Phe36 in Chai-1 imply distances shorter than the sum of the van der Waals radii, indicating pose-dependent clashes (Fig. 5c). Despite coronene being positioned in the canonical pocket, neither model accommodates the ligand through backbone shifts or pocket expansion to resolve clashes.

GOBP2 BLAST analysis revealed low sequence identity (30-35%) and high query coverage (>85%) across all aligned sequences, all seven OBPs (Fig. 5d). This illustrates the high sequence diversity of OBPs but the conserved functional and structural characteristics. Discontiguous BLAST revealed only seven OBPs, reflecting the limited characterisation and sequence heterogeneity of this family.

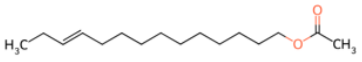
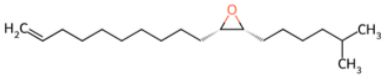
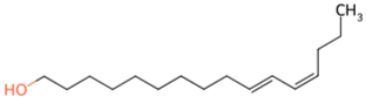
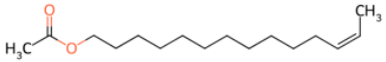
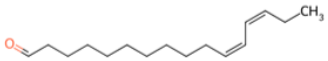
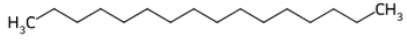
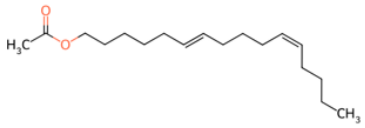
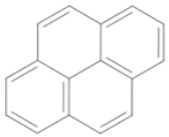
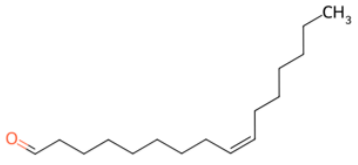
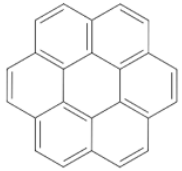
For each OBP, both native and non-native ligands (from aligned OBPs) were predicted alongside coronene, pyrene and hexadecane (Table 2). Native ligands typically represent species-specific pheromones, whereas non-native ligands serve as

chemically and sterically implausible controls to evaluate whether Chai-1 and Boltz-2 can distinguish biologically implausible binding.

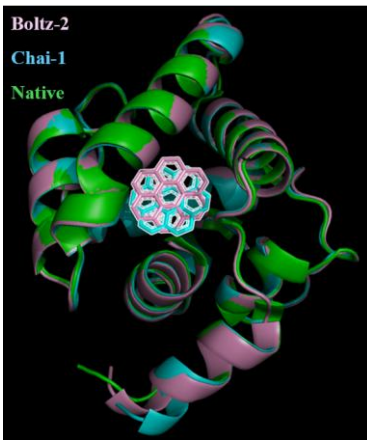
Analysis of *Amyelois transitella* PBP1 structures places the native ligand (11Z,13Z)-hexadecadienal within the central binding pocket, forming energetically favourable interactions with the surrounding residues (Fig. 5e). Boltz-2 and Chai-1 MSA-based predictions reproduce binding within the central pocket, forming extended, U-shaped conformations (Fig. 5f, g). The aldehyde tail of (11Z,13Z)-hexadecadienal clashes with Ile52 in both models, with Boltz-2 showing extensive overlap and hydrophobic contacts, suggesting a shift in ligand pose compared to Chai-1. Van der Waals contacts were consistently observed with Phe12, Leu8, and Ile94, while Chai-1 uniquely predicts additional contacts with the Ser9 backbone.

For the non-native ligand 2-methyl-7R,8S-epoxy-17-octadecene, both models predict binding within the canonical hydrophobic pocket in an extended conformation (Fig. 5h, i). Predicted van der Waals interactions include Ile52, Val62, Phe12, Ser9, Leu8, and Ile94. Boltz-2 shows significant steric overlap with the Ile94 side chain, with minor overlaps with Ser9 backbone and Leu8 side chains (Fig. 5h). Chai-1 predicted small overlaps with Ile94 and Val62 side chains (Fig. 5i). The epoxy group showed minimal steric conflicts in both models. These results suggest that although both models accommodate this non-native ligand within the pocket, consistent steric clashes at the methyl terminus (Ile93) highlight potential inaccuracies, with models overestimating chemical and steric compatibility of non-native compounds.

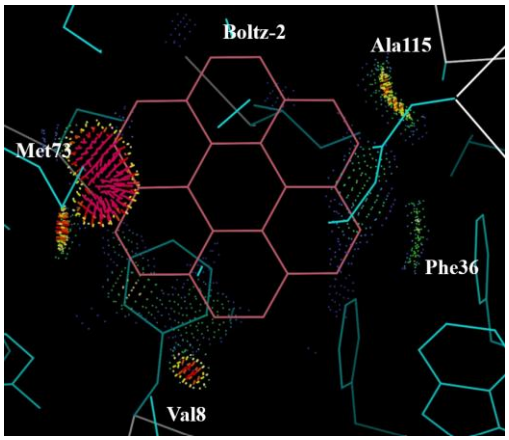
Table 2. Native and Non-Native Ligands for BLAST-aligned sequences to *Apis mellifera* Pheromone Binding Protein 1 (ASP1)

	Organism	OBP	PDB	Ligand	Structural Formula		Organism	OBP	PDB	Ligand	Structural Formula
1	*Epiphyas postvittana*	PBP3	6VQ5	(Z)-11-tetradecenyl acetate		6	*Lymantira dispar*	LdisPBP1	6UM9	2-methyl-7R,8S-Epoxy-17-octadecene	
2	*Bombyx mori*	BmPBP	1XFR	Bombykol		7	*Ostrinia furnacalis*	PBP2	7UO6	Z-12-tetradecenyl acetate	
3	*Amyelois transitella*	PBP1	4INW	(11Z,13Z)-hexadecadienal		8	-	-	-	Hexadecane	
4	*Antheraea polyphemus*	ApolPBP	1QWV	6E,11Z-hexadecadienyl acetate		9	-	-	-	Pyrene	
5	*Helicoverpa armigera*	PBP1	7VW8	(Z)-hexadec-9-enal		10	-	-	-	Coronene	

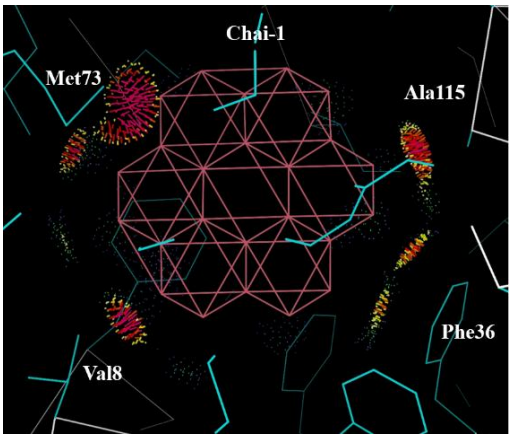
a)



b)



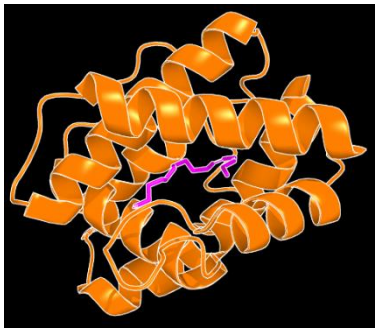
c)



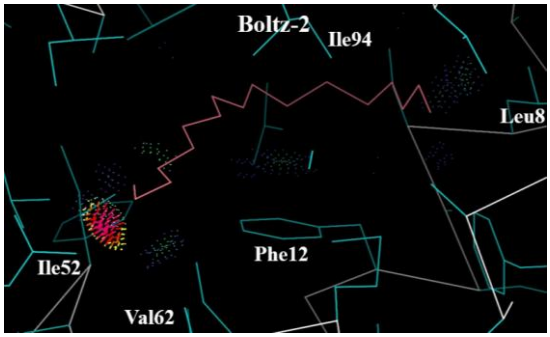
d)

Name	Organism	PDB code	Query Cover	% Identity	Native Ligand
PBP3	*Epiphyas postvittana*	6VQ5	95	35	(Z)-11-tetradecenyl acetate
BmPBP	*Bombyx mori*	1XFR	88	35	Bombykol
PBP1	***Amyelois transitella***	**4INW**	**95**	**34**	**(11Z,13Z)-hexadecadienal**
ApolPBP	*Antheraea polyphemus*	1QWV	95	31	6E,11Z-hexadecadienyl acetate
PBP1	*Helicoverpa armigera*	7VW8	95	31	(Z)-hexadec-9-enal
LdisPBP1	*Lymantira dispar*	6UM9	93	30	2-methyl-7R,8S-Epoxy-17-octadecene
PBP2	*Ostrinia furnacalis*	7UO6	90	32	Z-12-tetradecenyl acetate

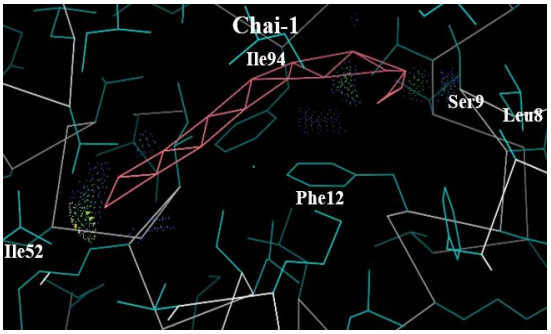
e)



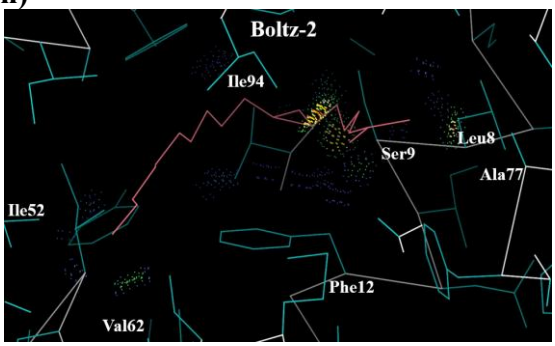
f)



g)



h)



i)

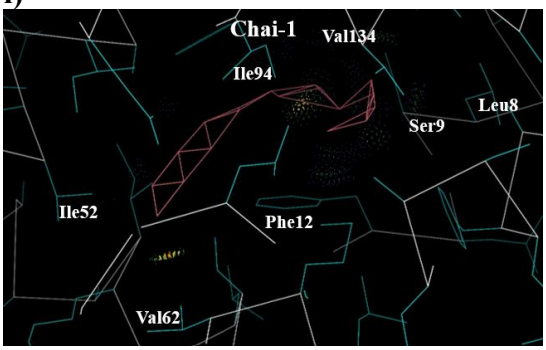


Figure 5. Predicted Van der Waals contacts between ligands and side chains for Boltz-2 and Chai-1 MSA-based predictions. Contacts are represented by blue/green dots, with steric overlaps indicated in red/yellow. Side chains are depicted in blue and Cα atoms in white. Residues with Van der Waals contacts, steric or hydrophobic clashes are labelled. **(a)** Predicted Coronene binding within the hydrophobic pocket of *Bombyx mori* GOBP2 **(b-c)** Van der Waals overlaps for Boltz-2 and Chai-1 Coronene predictions. **(d)** BLAST-based *Bombyx mori* GOBP2 overlaps aligned to homologs with high query coverage but low identity. **(e)** Canonical binding of *Amyelois transitella* PBP1 to the native ligand (11Z,13Z)-hexadecadienal. **(f-g)** Van der Waals overlaps for Boltz-2 and Chai-1 (11Z,13Z)-hexadecadienal. **(h-i)** Van der Waals overlaps for Boltz-2 and Chai-1 2-methyl-7R,8S-epoxy-17-octadecene.

In silico-mutated Odorant Binding Protein bound to non-native Hexadecane.

Predicted complexes of an in silico-mutated *Bombyx mori* GOBP2 bound to the non-native ligand hexadecane were generated using Chai-1 and Boltz-2, with and without MSA/template inputs. Canonical interacting residues were mutated to Glycine to remove side-chain-mediated interactions (Fig. 6a).

Boltz-2 predictions struggled to model plausible hexadecane binding. The ligand collapsed into compact, chemically unrealistic conformations, reducing the number of interactions within 4Å to five residues (Fig. 6b, c). Both inputs place hexadecane within a central binding pocket. With contextual information, Boltz-2 predicts a canonical OBP-like fold with 8 α-helices, but without input, it produces a mixture of α-helices and β-sheets. Predicted interacting residues include Phe12 and Leu48, hydrophobic residues expected to only form weak van der Waals contacts with hexadecane. Despite the chemical implausibility, Boltz-2 predicted high binding affinities (~2 μM) with and without templates but predicted low binding probabilities (~0.10).

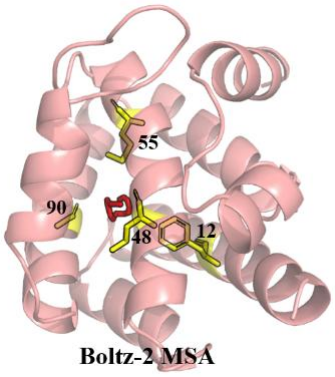
Conversely, Chai-1 retained chemically plausible ligand poses and conformations. With MSA, hexadecane adopted a U-shaped conformation, whereas without it remained extended and linear (Fig. 6d, e). Both predicted an OBP-like fold, with 8 α-helices, and placed hexadecane within a central binding pocket. Mutated Glycines appeared frequently among predicted contacts (MSA: Gly8, Gly37, Gly115, Gly118; non-MSA: Gly90, Gly94, Gly118), suggesting Chai-1 may retain contextual information from the original hydrophobic residues through MSA inputs or residue-level embeddings (Fig. 6a). Limited overlap of interacting residues was observed between Boltz-2 and Chai-1, with only Phe12 shared between Boltz-2 MSA/non-MSA and Chai-1 non-MSA predictions. Some residue overlap was observed across the two Chai-1 predictions (Trp37, Gly118, Ser122).

Both models predicted binding of chemically implausible hexadecane within the mutated protein. Boltz-2 produced structurally inconsistent and unrealistic ligand poses, whereas Chai-1 maintained an OBP-like backbone and more chemically plausible binding conformations, even in the absence of MSA inputs. Both modelled hexadecane binding within the central binding pocket, but only Chai-1 produced reasonable ligand poses.

a)

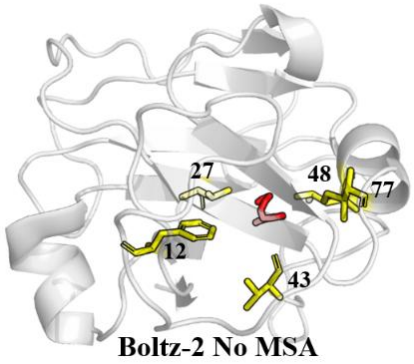
```
                    0         10        20        30        40        50        60        70
GOBP2 (2WCK - PDB)  TAEVMSHVTAHFGKTLEECREESGLSVDILDEFKHFWSDDFDVVHRELGCAIICMSNKFSLMDDDVRMHH
Altered GOBP2       TAEVMSHGTAHFGKTLEECREESGLSVDILDEGKHGWSDDFDVVHRELGCAGICMSNKFSLGDDDGGGHH
Boltz-2 MSA         TAEVMSHGTAHFGKTLEECREESGLSVDILDEGKHGWSDDFDVVHRELGCAGICMSNKFSLGDDDGGGHH
Boltz-2 NO MSA      TAEVMSHGTAHFGKTLEECREESGLSVDILDEGKHGWSDDFDVVHRELGCAGICMSNKFSLGDDDGGGHH
Chai-1 MSA          TAEVMSHGTAHFGKTLEECREESGLSVDILDEGKHGWSDDFDVVHRELGCAGICMSNKFSLGDDDGGGHH
Chai-1 NO MSA       TAEVMSHGTAHFGKTLEECREESGLSVDILDEGKHGWSDDFDVVHRELGCAGICMSNKFSLGDDDGGGHH
                              80        90        100       110       120       130       140
GOBP2 (2WCK - PDB)  VNMDEYIKSFPNGQVLAEKMVKLIHNCEKQFDTETDDCTRVVKVAACFKEDSRKEGIAPEVAMVEAVIEKY
Altered GOBP2       VNMDEYIKSFPNGQVLAEKGVKLGHNCGKQFDTETDDCTGVVKGGACGKEDSRKEGIAPEVAMVEAVIEKY
Boltz-2 MSA         VNMDEYIKSFPNGQVLAEKGVKLGHNCGKQFDTETDDCTGVVKGGACGKEDSRKEGIAPEVAMVEAVIEKY
Boltz-2 NO MSA      VNMDEYIKSFPNGQVLAEKGVKLGHNCGKQFDTETDDCTGVVKGGACGKEDSRKEGIAPEVAMVEAVIEKY
Chai-1 MSA          VNMDEYIKSFPNGQVLAEKGVKLGHNCGKQFDTETDDCTGVVKGGACGKEDSRKEGIAPEVAMVEAVIEKY
Chai-1 NO MSA       VNMDEYIKSFPNGQVLAEKGVKLGHNCGKQFDTETDDCTGVVKGGACGKEDSRKEGIAPEVAMVEAVIEKY
```

b)



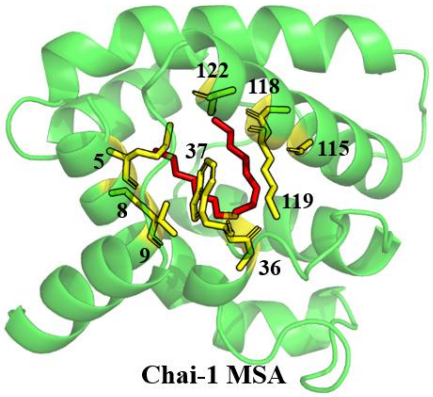
Boltz-2 Predicted Affinity (μM): 2.27
Boltz-2 Probability Binary: 0.10

c)



Boltz-2 Predicted Affinity (μM): 1.78
Boltz-2 Probability Binary: 0.11

d)



e)

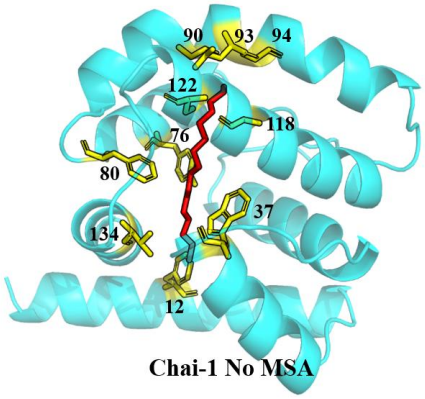


Figure 6. In silico mutated *Bombyx mori* General Odorant-Binding Protein 2 (GOBP2) bound to non-native hexadecane. Glycine substitutions were introduced to remove side-chain-mediated interactions within 4Å of the ligand from the canonical binding pocket residues (yellow), allowing for comparison with the non-polar ligand hexadecane (red). Predicted binding affinity (μM) and binary probability outputs (0-1) are only provided by Boltz-2. **(a)** Mutated GOBP2 sequences: canonical interacting residues and their glycine substitutions are highlighted in yellow, model-predicted hexadecane contacts are highlighted in blue. **(b)** Boltz-2 prediction with MSA and template input, showing predicted binding affinity and binary probability. **(c)** Boltz-2 prediction without MSA/template. **(d)** Chai-1 prediction with MSA/template. **(e)** Chai-1 prediction without MSA/template.

Discussion and Conclusion

Our analysis of Chai-1 and Boltz-2 reveals a consistent separation between backbone prediction and ligand placement. Whilst both models accurately reproduce OBP backbone folds when contextual information such as MSAs and templates is available, their ligand poses are frequently sterically or chemically implausible, failing to reflect realistic protein-ligand interactions. Odorant binding proteins (OBPs) were chosen as a test set due to their high structural conservation, limited sequence conservation, scarcity of contextual information, and binding promiscuity (Fig. 1) [62-69].

Chai-1 and Boltz-2 were selected for their accessibility, available either locally without high computational requirements or via proprietary web servers [82, 86]. Both are based on and developed from AlphaFold neural network architecture, with Boltz-2 predicting both protein-ligand structure and affinity, while Chai-1 uniquely integrates an embedded protein language model to guide backbone prediction and subsequent ligand placement [18, 22, 29, 42, 43, 89]. Both models generated accurate OBP backbones when supplied with contextual information, closely matching crystallised experimental structures (Fig. 2, 3). Chai-1 maintained high backbone accuracy without MSA and template input, whereas Boltz-2 relied heavily on MSAs and templates to reproduce the conserved α-helical structure of OBPs.

Despite accurate backbone modelling with contextual information, ligand placement was consistently poor for both models. Ligands were positioned in canonical hydrophobic pockets without considering steric clashes, van der Waals overlaps, and chemical plausibility. Boltz-2 frequently mispositioned ligand poses, interacting residues, and hallucinated binding sites relative to crystallographic data, even with MSA and template information. Under non-MSA conditions or with mutated backbones, Boltz-2 performance further degraded, with distorted backbone structures and implausible ligand placement (Fig. 6). Chai-1 also placed native and non-native ligands within canonical pockets, but ignored van der Waals overlaps, steric clashes, and side-chain context, appearing to rely on structural priors rather than responding to mutations or ligand diversity.

Both models' predictive methods appear limited, with backbone modelling resembling traditional homology-based approaches, reliant on MSAs, templates, or embedded protein language models, whilst ligand docking appears naïve [23-31]. Ligand binding predictions appear simplistic, with docking reduced to the placement of ligands within predefined pockets, neglecting key biophysical constraints, including steric compatibility, hydrophobic interactions, and chemical plausibility.

Backbone Modelling as a Foundation

Accurate backbone modelling underpins accurate Chai-1 and Boltz-2 protein-ligand binding predictions, making it a critical test for model reliability. With MSAs and template inputs, both models reproduce OBP folds closely matching crystallised structures, with low RMSD(Å) (<1) values and high template modelling scores (>0.9) (Fig. 4a, b, c, d). However, model behaviour diverges sharply without contextual input; Chai-1 maintains a robust OBP backbone fold prediction, whereas Boltz-2 predictions collapse. Backbone predictions were only accurate when MSAs, templates, or Chai-1's embedded protein language model was available, mirroring traditional homology-modelling techniques.

Boltz-2 showed a strong bias towards homology modelling. With MSAs and templates available, it produced plausible α-helical backbones, but without them, predictions degrade into mixtures of α-helices and implausible β-sheets. Even under favourable conditions, Boltz-2 can inaccurately model the orientation and placement of α-helical domains, subsequently shifting hydrophobic binding pockets and distorting ligand placement (Fig. 3d, c). These disruptions lead to few interacting residues overlapping between crystallised and predictive structures. Reliance on evolutionary information makes Boltz-2 unsuitable for modelling proteins with shallow MSAs, and predictions should be interpreted cautiously.

Chai-1 retains accurate domain folds regardless of input, reflecting the strength of its integrated protein language model. Backbone accuracy remains consistent even with synthetic mutations, retaining OBP-like α-helical backbones. This suggests Chai-1's embedded protein language model dominates backbone predictions, with MSA and templates having little overall impact on the overall fold. Binding pockets appear correctly positioned but likely reflect structural priors rather than genuine chemical plausibility.

OBPs with shallow MSAs and poor characterisation highlight both models' weaknesses. Boltz-2's reliance on MSA/template information and Chai-1's reliance on its embedded protein language model undermines ligand predictions, especially for orphan or understudied protein families [48]. For Chai-1, this can lead to overgeneralisation of protein folds and binding poses, leading to more subtle variations being obscured, while Boltz-2 may hallucinate binding pockets completely. [116, 117].

The limitations of both models are partly obscured by CASP-style benchmarks, which reward static, domain-level accuracy rather than domain orientation or inter-domain packing [118]. Metrics such as the Global Distance Test-Total Score (GDTTS) favour models which can reconstruct individual domains well, even if orientation is incorrect [119]. Chai-1 and Boltz-2 achieve high scores for local backbone accuracy but struggle with inter-domain metrics such as iPTM (Fig. 4e, f). While optimisation towards CASP-

style evaluation highlights static backbone accuracy, it limits applicability for modelling protein flexibility and dynamics, affecting their ability to capture downstream ligand binding errors and accurately represent residue-level interactions.

Protein-Ligand Binding Predictions

Both models predicted binding of sterically and chemically implausible ligands such as coronene, a large, inflexible, polycyclic hydrocarbon. Boltz-2 and Chai-1 placed coronene within the central hydrophobic binding pocket of *Bombyx mori* GOBP2 with limited variation in pose (Fig. 5a). KiNG analysis showed extensive steric clashes, with neither model adjusting their backbone to accommodate coronene binding and facilitate its entry through the single, narrow entrance of GOBP2 (Fig. 5b, c). This reinforces the idea that the homology-based backbone modelling occurs first before the ligands are subsequently positioned according to structural priors rather than physical feasibility.

In silico mutation of *Bombyx mori* GOBP2, substituting canonical interacting residues with glycine, caused Boltz-2 predictions to collapse, producing irregular ligand poses and introducing non-canonical β-sheets without MSA input (Fig. 6a, b, c). Chai-1, by contrast, retained canonical OBP structure, with hexadecane adopting biologically plausible poses similar to known molecules. Chai-1 often positioned ligands relative to the original sequence positions of hydrophobic residues, suggesting reliance on learned structural priors rather than actively responding to the mutated sequence and non-polar glycine residues (Fig. 6a, e, f).

Uniquely, Boltz-2 returns binding probability binaries and predicted affinities for protein-ligand complexes; however, these are inconsistent with literature values and chemical reality [70-79, 120-122]. With and without MSA inputs, Boltz-2 frequently predicted low probability binding binaries with μM-range affinities, contradicting nanomolar affinities observed in vitro and in vivo (Fig. 4g, h). Even for the in silico mutated *Bmor*GOBP2 bound to the hexadecane molecule, Boltz-2 predicted strong binding affinities (MSA: 2.27 μM, non-MSA: 1.78 μM) alongside low binding probabilities (MSA: 0.10, Non-MSA:0.11). These inconsistencies indicate that Boltz-2's free-energy perturbation-based (FEP) affinity module is poorly calibrated and may assume oversaturated ligand environments, with predictions reflecting energetically optimal poses unrepresentative of biological reality. Ongoing development of the Boltz models, including Boltz-ABFE, may address these limitations [123].

Whilst Boltz-2's affinity and probability outputs are framed as unique features, they are poorly integrated into the core modelling framework and do not reflect chemical or biological reality. Accurate backbone modelling is a prerequisite for meaningful ligand poses, but current models show persistent issues with steric clashes, pose diversity,

and non-native ligands, and as a result, render affinity predictions misleading. They offer a superficial sense of quantitative precision but fail to capture biological or physical reality. In practice, non-native ligands should return zero binding probability, with affinities far below that of native ligands, whereas native ligands should return experimentally verified nanomolar affinities. Combined, these findings suggest Chai-1 and Boltz-2 are best suited for generating structural hypotheses or providing crystallisation support for well-characterised proteins, applications that fit within traditional drug discovery or exploratory pipelines, rather than as credible predictors of ligand binding or affinity [124-126].

Limitations in Current Model Architecture

Our results highlight that Chai-1 and Boltz-2 generate protein backbones independently of ligand placement, leading to steric clashes and accommodation of chemically implausible ligands (Fig. 6, 7). This decoupling reflects their AlphaFold-derived "block" architectures, which optimise for local backbone accuracy [18, 22, 82, 86]. As a result, ligand positions appear to be guided by structural priors rather than explicit energetic feasibility and global context. This reproduces superficially correct structures but fails to capture ligand-induced flexibility.

This architecture reflects a deliberate trade-off: by siloing backbone and affinity modules, models maintain stability and perform strongly on structure-focused benchmarks like CASP, but at the cost of modelling realistic ligand interactions [118]. Models may focus on molecule positioning whilst overlooking the prediction of precise protein-ligand interactions [127]. Missing this critical information may lead to a gap between academic benchmarks and real-world drug design, a key application of such models. As such, predictions may appear credible using static evaluations but cannot be relied upon for modelling physically coherent binding events, particularly in demanding scenarios with sparse evolutionary information.

From a broader perspective, such issues reflect issues of deep learning systems within structural biology and our inability to understand how "black-box" models make such decisions [128, 129]. Models encode residue-residue and protein-ligand interactions in high-dimensional learned representations, which capture biologically relevant relationships but are not readily interpretable [22]. Without control mechanisms to assess global steric and chemical plausibility, errors remain hidden, leading predictions to appear superficially correct but chemically nonsensical. For orphan proteins and poorly characterised protein families, those of most interest in therapeutic or exploratory contexts, sparse evolutionary information and over-reliance on MSA and template information may bias models towards reproducing known homology rather than predicting novel folds [130]. Addressing this will likely require architectural

changes that integrate chemical constraints without sacrificing backbone accuracy, placing limits on their immediate utility in fields like drug discovery and exploratory protein development.

Intellectual Property Challenges

Model limitations observed during this study, like ligand placements that appear superficially plausible and the dominance of backbone modelling, are compounded by the licensing and distribution of machine-learning models. Boltz-2 and Chai-1 are released under the permissive Apache 2.0 licence, allowing commercial use, modification, and redistribution [82, 86, 131]. By contrast, AlphaFold3 has restricted access, with full model parameters, weights, and inferences only available at DeepMind's discretion for non-commercial use and are protected by a broad range of patents covering the use of MSAs, pairwise representations, and use of transformer-like mechanisms for predicting protein structure [22, 132, 133]. This creates two differing methods of machine learning protein prediction infrastructure: open with permissive development versus closed with proprietary control.

Open-source licensing promotes rapid uptake and iteration. Open-source models like Chai-1 and Boltz-2 are capable of being iterated upon by commercial ventures and academic institutions using proprietary datasets. Yet improvements, especially those made in commercial settings, often remain unpublished, reducing transparency. Closed-source models, such as AlphaFold3, attract significant investment, with DeepMind's commercial spinout, Isomorphic Labs, raising $600 million in series A funding to accelerate drug discovery [134]. However, this comes at the cost of reproducibility and transparency, with independent verification limited.

Without transparency in either code or training data, chemically implausible predictions may remain unchallenged, undermining confidence in results. In drug discovery, a pertinent application of such models, this creates both ethical and practical risks. Whether open or closed, models risk becoming siloed and biased, limiting the scientific community's trust in their future utility, accuracy, and reproducibility. This exacerbates the purely scientific challenges highlighted by our findings. Further progress may be dependent on open-source model transparency and shared standards of data availability and training data.

Future directions

Future improvements to protein-ligand predictive models may come from integrating chemical and energetic priors directly into existing architectures to improve ligand binding without compromising backbone accuracy [103, 135-137]. Current pair

representations capture local residue-residue relationships, but with insufficient information flow between affinity output (pose, binding probability, and affinity) and backbone modules to enforce chemical realism at a global scale [18, 22]. Architectural modifications could address this by adding explicit van der Waals checks after initial complex generation to flag steric clashes and improve prediction [115]. Incorporation of molecular dynamic refinement and light post-processing may mitigate chemically implausible outcomes.

Iterative feedback loops could be integrated. Steric violations (e.g., “a carboxyl group from ligand 1 is overlapping with backbone Ala112”) are fed back into the representation layers. This would allow structural priors to reflect steric feasibility while maintaining the accuracy and stability of the current framework. However, adding architectural complexity will increase the already high computational requirements, and balancing these two factors is challenging.

Practical applications could include integration into traditional drug discovery pipelines. Predictive models could refine known antibody therapeutics by proposing small mutations to optimise binding interfaces in silico [138-142]. Large panels of mutated candidate binders could be rapidly screened computationally before narrowing them down to a small subset for experimental validation, reducing time and cost relative to iterative wet lab cloning and crystallisation. Hybrid workflows could allow effective discrimination of binders with desired characteristics before more targeted, wet-lab investigation [20].

Conclusion

Our analysis of Chai-1 and Boltz-2 reveals a clear divide between plausible backbone prediction and unreliable ligand placement. Contextual information, including MSAs, templates, and embedded protein language models, significantly improves backbone modelling, yet ligand predictions default to known prior poses and fail to respect steric and chemical feasibility. Using odorant binding proteins, a structurally conserved but sequentially diverse family of proteins that bind a wide range of ligands, we test both models under stress conditions with limited contextual information.

Chai-1, supported by its embedded protein language model, consistently reproduced OBP backbone folds with and without MSA and template information. Ligand placement within canonical binding pockets was superficially plausible, following contextual priors, but steric overlaps and unrealistic residue-ligand contacts persisted. Boltz-2, without an embedded protein language model, relies heavily on contextual MSA/template information, with predictions collapsing for both ligand binding and backbone prediction under stress conditions or after in silico mutation. Additionally, its

dedicated binding affinity module is poorly integrated, producing high-affinity predictions for implausible ligands and low probability binaries for native ligands.

These observations underscore that current architectures remain closer to advanced homology modelling with ligand superimposition than genuine protein-ligand simulation. Their current real-world applications for fields like drug discovery and protein characterisation are limited, though impressive developments and effective for generating interest and investment; they lack the necessary robust accuracy. At present, Chai-1 and Boltz-2 are best viewed as hypothesis-generating tools, providing crystallisation support for well-characterised protein families rather than reliable predictors of ligand binding or affinity [124-126].

Overall, this work highlights the impressive progress and infancy of machine-learning-based protein-ligand binding models. Backbone accuracy is advancing rapidly, with contextual information dominating the field, but chemically realistic ligand binding lags. Addressing this gap will require not only architectural innovation but new standards to ensure ligand predictions extend beyond superficial plausibility to applicable biological insights.

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